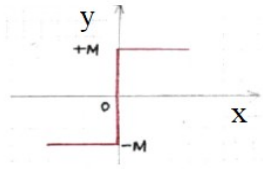


DRAFT SCHEME

1000EET401052301

Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
SEVENTH SEMESTER B. TECH (S) DEGREE EXAMINATION, MAY 2023			
Course Code: EET401			
Course Name: ADVANCED CONTROL SYSTEMS			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions, each carries 3 marks.	Marks
1		Any three advantages (1Mark Each)	(3)
2		Definition of Phase variable (1Mark), Any two advantages (1Mark Each)	(3)
3		Complete derivation (3 Marks) $\dot{X}=AX +BU, Y=CX+DU$ Taking Laplace Transform and initial conditions where zero $SX(S)=AX(S)+BU(S),$ $X(S)=[(SI-A)^{-1}B] U(S)$ $Y(S)=CX(S)+DU(S)=[C(SI-A)^{-1}B+D] U(S)$ $Y(S)/U(S)= [C(SI-A)^{-1}B+D]$	(3)
4		Three properties (1Mark Each)	(3)
5		Definition of Controllability(1Mark) Steps for controllability check using Kalman's test (2 Marks)	(3)
6		Duality principle- Explanation with necessary equations(3Marks)	(3)
7		Frequency entrainment explanation (2 Marks), Characteristic Curve (1 Mark)	(3)
8		Input signal: $x = X \cdot \sin \omega t$ Output signal: $y(\omega t) = \begin{cases} +M & ; 0 < \omega t < \pi \\ -M & ; \pi < \omega t < 2\pi \end{cases}$  Describing function : $K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1 \quad Y_1 = \sqrt{A_1^2 + B_1^2} \quad \phi_1 = \tan^{-1} \left(\frac{B_1}{A_1} \right)$ $A_1 = \frac{4M}{\pi} \quad B_1 = 0 \quad \phi_1 = 0^\circ$	(3)
9		Explanation with diagrams (3Marks)	(3)

DRAFT SCHEME

1000EET401052301

10		The middle matrix is found as the coeff. of $\begin{bmatrix} \frac{x_1^2}{2} & \frac{x_1x_2}{2} & \frac{x_1x_3}{2} \\ \frac{x_2x_1}{2} & \frac{x_2^2}{2} & \frac{x_2x_3}{2} \\ \frac{x_1x_3}{2} & \frac{x_2x_3}{2} & \frac{x_3^2}{2} \end{bmatrix} \quad (1 \text{ Mark})$ $x_1^2 + 3x_2^2 + 11x_3^2 - 2x_1x_2 + 4x_2x_3 + 2x_1x_3 = [x_1 \ x_2 \ x_3] \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & 2 & 11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ Now, $p_{11} = 1 > 0$ $\begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix} = 2 > 0$ $\begin{vmatrix} 1 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & 2 & 11 \end{vmatrix} = 11 > 0 \quad (1 \text{ Mark})$ Quadratic form is positive definite (1 Mark)	(3)
PART B			
<i>Answer one full question from each module, each carries 14 marks.</i>			
Module I			
11	a)	$x_1 = i_L, \quad x_2 = v_c \quad (1 \text{ Mark})$ $\frac{L di_L}{dt} + i_L R_2 = v_c$ $\dot{x}_1 = \frac{-10}{5} x_1 + \frac{v_c}{5} \quad (2 \text{ Marks})$ $i_L + C \frac{dv_c}{dt} = \frac{v_0 - v_c}{R_1}$ $\dot{x}_2 = \frac{(v_0 - v_c)5}{5} - 5x_1 \quad (2 \text{ Marks})$ $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 1/5 \\ -5 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad (2 \text{ Marks})$	(7)
	b)	State model of field-controlled dc motor: KVL equation for field circuit $v_f = R_f i_f + L_f \frac{di_f}{dt} \quad (1 \text{ mark})$ Torque equation $T \propto i_f, \quad T = K_{tf} i_f \quad (1 \text{ mark})$ Equation of electro mechanical system: $J \frac{d^2\theta}{dt^2} + B \frac{d\theta}{dt} = K_{tf} i_f \quad (1 \text{ mark})$ Selection of state variables and output variables. (1 mark) Substitution and rearrangement (1 mark) Final state equation (1 mark) Final output equation (1 mark)	(7)
OR			

DRAFT SCHEME

1000EET401052301

12	a)	$x_1 = y, x_2 = \dot{y} = \dot{x}_1, x_3 = \ddot{y} = \dot{x}_2$ (1 mark) $\dot{x}_1 = x_2$ (1 mark) $\dot{x}_2 = x_3$ (1 mark) $\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 10y = 8u(t).$ i.e. $\dot{x}_3 = 8u(t) - 10x_1 - 11x_2 - 6x_3$ (2 mark) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -10 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 8 \end{bmatrix} u$ (1mark) $y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ (1 mark)	(7)
	b)	$\frac{Y(s)}{U(s)} = \frac{2(s+5)}{(s+2)(s+3)(s+4)} = \frac{A}{(s+2)} + \frac{B}{(s+3)} + \frac{C}{(s+4)}$ (1Mark) A=3, B= -4, C=1 (3Marks) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u$ (2Marks) $y = \begin{bmatrix} 3 & -4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ (1Mark)	(7)
Module II			
13	a)	$(sI - A) = \begin{bmatrix} s & -1 \\ 6 & s+5 \end{bmatrix}$ (1 Mark) $e^{At} = L^{-1} [(sI - A)^{-1}] = \begin{bmatrix} 3e^{-2t} - 2e^{-3t} & e^{-2t} - e^{-3t} \\ -6e^{-2t} + 6e^{-3t} & -2e^{-2t} + 3e^{-3t} \end{bmatrix}$ (4 Marks) $X(t) = e^{At} X(0) = \begin{bmatrix} 3e^{-2t} - 2e^{-3t} & \\ -6e^{-2t} + 6e^{-3t} & \end{bmatrix}$ (2 Marks)	(7)
	b)	$\frac{Y(z)}{U(z)} = \frac{4z^3 - 12z^2 + 13z - 7}{z^3 - 4z^2 + 5z - 2}$	(7)

DRAFT SCHEME

1000EET401052301

		$x_1(k+1) = x_2$ $x_2(k+1) = x_3$ $x_3(k+1) = 2x_1 - 5x_2 + 4x_3 + u \text{ (2Marks)}$ $y = -7x_1 + 13x_2 - 12x_3 + 4x_3(k+1)$ $y = x_1 - 7x_2 + 4x_3 + 4u \text{ (2Marks)}$ $\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \text{ (2Marks)}$ $y = [1 \quad -7 \quad 4] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + 4u \text{ (1 mark)}$	
		OR	
14	a)	$[\lambda I - A] = \begin{bmatrix} \lambda & -1 \\ 2 & \lambda + 3 \end{bmatrix} \text{ (1 Mark)}$ $ \lambda I - A = 0, \lambda_1 = -1, \lambda_2 = -2 \text{ (1 Mark)}$ $\alpha_0 + \alpha_1 \lambda_1 = e^{\lambda_1 t} = e^{-t}$ $\alpha_0 + \alpha_1 \lambda_2 = e^{\lambda_2 t} = e^{-2t} \text{ (2 Mark)}$ $\alpha_1 = e^{-t} - e^{-2t}$ $\alpha_0 = 2e^{-t} - e^{-2t} \text{ (2 Mark)}$ $e^{At} = \alpha_0 I + \alpha_1 A \text{ (1 Mark)}$ $e^{At} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -2e^{-t} + 2e^{-2t} \end{bmatrix} \text{ (1 Mark)}$	(8)
	b)	$T(s) = C(SI - A)^{-1}B + D \text{ (1 Mark)}$ $(SI - A) = \begin{bmatrix} s+3 & -1 \\ 2 & s \end{bmatrix} \text{ (1 Mark)}$ $(SI - A)^{-1} = \begin{bmatrix} \frac{s}{s^2+3s+2} & \frac{1}{s^2+3s+2} \\ \frac{-2}{s^2+3s+2} & \frac{s+3}{s^2+3s+2} \end{bmatrix} \text{ (2 Marks)}$ $T(s) = [1 \quad 0] \begin{bmatrix} \frac{s}{s^2+3s+2} & \frac{1}{s^2+3s+2} \\ \frac{-2}{s^2+3s+2} & \frac{s+3}{s^2+3s+2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ (1 Mark)}$ $T(S) = \frac{1}{s^2+3s+2} \text{ (1 Mark)}$	(6)
		Module III	
15	a)	Necessary and sufficient condition for complete observability is none of the columns of $\tilde{C}$ matrix is zero $\tilde{C} = CM$ (1 Mark)	(10)

DRAFT SCHEME

1000EET401052301

		$ \lambda I - A = 0, \lambda_1 = -1, \lambda_2 = -1, \lambda_3 = -4$ (2 Marks) $M = \begin{bmatrix} 4 & 4 & 1 \\ -4 & -2 & 2 \\ -4 & 0 & -4 \end{bmatrix}$ (4 Marks) $\tilde{C} = CM = [1 \ 0 \ 0] \begin{bmatrix} 4 & 4 & 1 \\ -4 & -2 & 2 \\ -4 & 0 & -4 \end{bmatrix} = [4 \ 4 \ 1]$ (2 Marks) None of the columns of $\tilde{C}$ matrix is zero. Hence the system is completely observable. (1 Mark)	
	b)	Explanation of Full order observer(2Marks), Reduced order observer(2Marks)	(4)
		OR	
16	a)	$\lambda_1 = -1, \lambda_2 = -3$ $\text{Rank of } [(\lambda I - A) \ B] = 2$ (full rank) for both λ_1 and λ_2 Hence system is completely controllable (4 Marks) $\text{Rank of } \begin{bmatrix} C \\ (\lambda I - A) \end{bmatrix} = 2$ (full rank) for both λ_1 and λ_2 Hence system is completely observable (4 Marks)	(8)
	b)	Pole placement technique explanation (6Marks)	(6)
		Module IV	
17	a)	Definition of describing function (2 Marks) Stability Analysis- Concept of Enclosure of $-1/K_N$ locus by $G(j\omega)$ locus with necessary diagrams (2 Marks) Occurrence of Limit Cycle (2Marks)	(6)
	b)	Derivation of describing function of relay with dead zone with all necessary steps Waveforms(2Mark) Output Equation (1 Mark) Derivation (3 Marks) Final Answer (2 Marks)	(8)
		OR	
18	a)	Explanation of any three non-linearities with waveforms (2 Marks each)	(6)
	b)	Derivation of describing function of saturation nonlinearity with necessary steps Waveforms(2Mark) Output Equation (1 Mark)	(8)

DRAFT SCHEME

1000EET401052301

		Derivation (3 Marks) Final Answer (2 Marks)	
		Module V	
19	a)	Definition of Singular point (1 Mark), Any five types of singular point with nature of eigen values or phase portrait diagram (1 Mark each)	(6)
	b)	Lyapunovs Stability Criterion (3 Marks) $\dot{x}_1 = -3x_1 + x_2$ $\dot{x}_2 = -x_1 - x_2 - x_2^3$ $V(x) = x_1^2 + x_2^2$ $\dot{V}(x) = \frac{\partial V}{\partial x_1} \dot{x}_1 + \frac{\partial V}{\partial x_2} \dot{x}_2$ $\dot{V}(x) = 2x_1 \dot{x}_1 + 2x_2 \dot{x}_2 \quad (2 \text{ Marks})$ Substitute the values of $\dot{x}_1$ and $\dot{x}_2$ $\dot{V}(x) = 2x_1(-3x_1 + x_2) + 2x_2(-x_1 - x_2 - x_2^3)$ $\dot{V}(x) = -6x_1^2 - 2x_2^2 - 2x_2^4 = \text{negative definite} \quad (2 \text{ Marks})$ System is asymptotically stable in the large. (1 Mark)	(8)
		OR	
20	a)	$A^T P + P A = -Q$ $\begin{bmatrix} -1 & 1 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \quad (2 \text{ Marks})$ Solve for matrix P $P = \begin{bmatrix} 23/60 & -7/60 \\ -7/60 & 11/60 \end{bmatrix} \quad (5 \text{ Marks})$ Applying Sylvester's theorem prove that matrix P is positive definite. (2 Marks) Equilibrium state is stable (1 Mark)	(10)
	b)	Definition of Stability (1 Mark), Asymptotic stability (1 Mark), Asymptotically stable in the large (1 Mark), Instability. (1 Mark),	(4)
